Mathematics

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Contents

Chapter 1

Algebra

1.1 Groups

1.1.1 Magmas and monoids

Magmas and properties

Definition 1.1. A magma is a pair $(M, *)$ with M a set and

$$*: \ M \times M \to M$$

an internal composition law.

Definition 1.2. A magma $(M, *)$ is said to be associative if $\forall a, b, c \in M, \ (a*b)*c = a*(b*c)$.

Definition 1.3. A magma $(M, *)$ is said to be commutative if $\forall a, b \in M, \ a * b = b * a$.

Definition 1.4. An identity element of a given magma $(M, *)$ is an element $e \in M$, such that $\forall a \in M, \ e * a = a * e = a$.

Theorem 1.5. *If a magma $(M, *)$ has an identity element, then it is unique.*

Proof. If $e, e' \in M$ are identity elements for $*$, then $e = e * e' = e'$. □

Definition 1.6. Given a magma $(M, *)$ with an identity element $e \in M$. An element $a \in M$ is said to be invertible if there exists $b \in M$ such that $a * b = b * a = e$. b is then called an inverse of a.

Products of magmas

Definition 1.7. Given two magmas $(M, *)$ and $(N, \bullet)$, we define a canonical structure of magma on the set $M \times N$, defined by the following internal composition law:

$$\big((a, b), (a', b')\big) \mapsto (a * a', b \bullet b')$$

Proposition 1.8. *Let M and N be two magmas.*

(i) If M and N are associative (resp. commutative), then $M \times N$ is associative (resp. commutative),

(ii) If M and N have identity elements, then $M \times N$ has an identity element.

Proof. Admitted. □

Proposition 1.9. *Let M and N be two magmas with identity elements. If $a \in M$ and $b \in N$ are invertible, then $(a, b) \in M \times N$ is invertible.*

Proof. Admitted. □

Remark 1.10. The previous construction can be extended to any product of magmas. The propositions (1.8) and (1.9) then remain true.

Morphism of magmas

Definition 1.11. A morphism of magmas between $(M, *)$ and $(N, \bullet)$ is a map $f : M \to N$ such that

$$\forall a, b \in M, \ f(a * b) = f(a) \bullet f(b)$$

Proposition 1.12. *A composition of morphisms of magmas is a morphism of magmas.*

Proof. Admitted. □

Monoids and properties

Definition 1.13. A monoid is an associative magma with an identity element.

Theorem 1.14. *If $(M, *)$ is a monoid, then every invertible element $a \in M$ has a unique inverse, denoted by a^{-1}.*

Proof. Let $b, b' \in M$ be two inverses of a. Then we have

$$b = b * (a * b') = (b * a) * b' = b'.$$

□

Notation 1.15. If $(M, *)$ is a monoid, then for all $a \in M$, we denote:

$$a^0 = e \quad \text{and} \quad a^n = \underbrace{a * \cdots * a}_{n \text{ times}} \quad (\text{for } n \geq 1).$$

Proposition 1.16. *Given a monoid $(M, *)$, and $a, x, y \in M$,*

(i) If $a * x = a * y$ and a is invertible, then $x = y$,

(ii) If x is invertible, then x^{-1} is invertible and $(x^{-1})^{-1} = x$.

(iii) If x and y are invertible, then $x * y$ is also invertible and $(x * y)^{-1} = y^{-1} * x^{-1}$,

(iv) If x is invertible, then for all $n \in \mathbb{N},\ x^n$ is also invertible and $(x^n)^{-1} = (x^{-1})^n$. Hence we can denote $x^{-n} = (x^n)^{-1}$,

Proof. (i) $x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = y$.

(ii) Direct consequence of the invertible element definition.

(iii) $(x * y) * (y^{-1} * x^{-1}) = x * y * y^{-1} * x^{-1} = e$, and similarly for the other way.

(iv) It suffices to combine the two preceding properties in a proof by induction.

□

Proposition 1.17. *Any product of monoids is a monoid.*

Proof. It is induced by (1.8). □

Definition 1.18. A morphism of monoids is a morphism between underlying magmas, such that it preserves the identity element.

1.1.2 Groups

Groups and subgroups

Definition 1.19. A group is a monoid such that every element is invertible.

Proposition 1.20. *Any product of groups is a group.*

Proof. It is induced by (1.17) and (1.9). □

Definition 1.21. Let $(G, *)$ be a group. A subset $H \subseteq G$ is said to be a *subgroup* of G if:

(i) The identity element $e \in H$,

(ii) H is closed under the composition law, i.e., $\forall x, y \in H,\ x * y \in H$,

(iii) H is closed under taking inverses, i.e., $\forall x \in H,\ x^{-1} \in H$.

Proposition 1.22. *Let $(G, *)$ be a group. A non-empty subset $H \subseteq G$ is a subgroup of G if and only if:*

$$\forall x, y \in H, \quad x * y^{-1} \in H.$$

Proof. Suppose H is a subgroup of G. Let $x, y \in H$. By condition (iii) of Definition 1.21, $y^{-1} \in H$. Then, by condition (ii), $x * y^{-1} \in H$.
Conversely, suppose $H \neq \emptyset$ and satisfies the given criterion. Since H is non-empty, there exists some $x \in H$. Applying the criterion to x and x yields $x * x^{-1} = e \in H$, which satisfies condition (i). Now, for any $y \in H$, applying the criterion to e and y gives $e * y^{-1} = y^{-1} \in H$, satisfying condition (iii). Finally, for any $x, y \in H$, we know $y^{-1} \in H$, so applying the criterion to x and y^{-1} gives $x * (y^{-1})^{-1} = x * y \in H$, satisfying condition (ii). □

Proposition 1.23. *Given a group G, any intersection of subgroups of G is a subgroup of G.*

Proof. Admitted. $\square$

Morphism of groups

Definition 1.24. A morphism of groups is a morphism of underlying magmas.

Definition 1.25. Let $f : G \to G'$ be a morphism of groups. We say that G is the domain of f, and G' is the codomain of f.

Proposition 1.26. *Let G, G' be two groups and $f : G \to G'$ be a morphism of groups. Then $f(e_G) = e_{G'}$ and for all $x \in G$, $f(x^{-1}) = f(x)^{-1}$.*

Proof. Admitted. $\square$

Proposition 1.27. *A composition of morphisms of groups is a morphism of groups.*

Proof. Induced by (1.12). $\square$

Theorem 1.28. *Let G, G' be two groups and $f : G \to G'$ be a morphism of groups.*

(i) If H is a subgroup of G, then $f(H)$ is a subgroup of G',

(ii) If H' is a subgroup of G', then $f^{-1}(H')$ is a subgroup of G.

Proof. Admitted. $\square$

Definition 1.29. The kernel of a morphism of groups $f : G \to G'$ is defined by

$$\mathrm{Ker}(f) = \{a \in G \mid f(a) = e_{G'}\}$$

Proposition 1.30. *The kernel of a morphism of groups $f : G \to G'$ is a subgroup of G.*

Proof. Induced by (1.28)-(ii). $\square$

Proposition 1.31. *A morphism of groups $f : G \to G'$ is injective if and only if its kernel is the trivial subgroup $\{e_G\}$.*

Proof. The if condition: Let $x, x' \in G$ be such that $f(x) = f(x')$, which is equivalent to $f(x)f(x')^{-1} = e_{G'}$. However, $f(x)f(x')^{-1} = f(x)f((x')^{-1}) = f(x(x')^{-1})$. Which implies, by hypothesis $x(x')^{-1} = e_G$. So $x = x'$. The reverse is obvious. $\square$

Definition 1.32. The image of a morphism of groups $f : G \to G'$ is defined by

$$\mathrm{Im}(f) = \{b \in G' \mid \exists a \in G,\ f(a) = b\}$$

Proposition 1.33. *The image of a morphism of groups $f : G \to G'$ is a subgroup of G'.*

Proof. Induced by (1.28)-(i). $\square$

Isomorphism of groups

Definition 1.34. An isomorphism of groups is a bijective morphism of groups.

Notation 1.35. If $f : G \to G'$ is an isomorphism, then we say that G is isomorphic to G'. If $G = G'$, we say that f is an automorphism.

Theorem 1.36. *The composition of two isomorphisms of groups is an isomorphism of groups.*

Proof. Obvious. □

Theorem 1.37. *If f is an isomorphism of groups, then the inverse f^{-1} is a morphism of groups. Moreover, it is an isomorphism of groups.*

Proof. Let $f : G \to G'$ be an isomorphism of groups and $y, y' \in G'$. Then

$$\begin{aligned} f^{-1}(yy') &= f^{-1}\Big(f\big(f^{-1}(y)\big)f\big(f^{-1}(y')\big)\Big) \\ &= f^{-1}\Big(f\big(f^{-1}(y)f^{-1}(y')\big)\Big) \\ &= f^{-1}(y)f^{-1}(y') \end{aligned}$$

□

Proposition 1.38. *The relation "is isomorphic to" is an equivalence relation on the class of groups. Its equivalence classes are called isomorphism classes.*

Proof. Induced by (1.37) and (1.36). □